PAVAN KUMAR MATHI

Software & AI Engineer | Python · LLM Agents · Computer Vision · Full-Stack Web

+91 6281394267 | mpk92789@gmail.com | Vizianagaram, Andhra Pradesh | linkedin.com/in/pavan-kumar-mathi-b554683a | github.com/pavankumarmathi489-cpu | pavankumarmathi489-cpu.github.io/portfolio

SUMMARY

Final-year B.Tech ECE student (2027) who builds AI software that runs offline on real devices. Built KAISER, a local AI agent with 41 tools and 365 automated tests; trained a YOLOv8 detector (mAP@50 0.951) for an edge-AI rover; and shipped a live Flask e-commerce site for a local business. Seeking an entry-level software, AI/ML or full-stack developer role.

TECHNICAL SKILLS

Languages: Python, C, JavaScript, SQL, HTML/CSS, Rust (basics)
AI / ML: LLM agents and tool calling, RAG and embeddings, LoRA fine-tuning, Ollama, PyTorch, Hugging Face Transformers, YOLOv8, OpenCV, MediaPipe, Whisper / Piper speech
Web & Backend: Flask, SQLAlchemy, PostgreSQL (Neon), REST APIs, Razorpay payments and webhooks, Jinja, Streamlit
Tools: Git, GitHub, Vercel, Linux (Debian), Playwright, Google Colab, n8n, Airtable, PuLP

PROJECTS

KAISER – Offline Personal AI Agent | *Python, Ollama, RAG, Whisper, Playwright*

- Built a voice-activated ("Hey Kaiser") agent that plans a task, asks approval for risky steps, runs it with 41 schema-defined tools and verifies the result, entirely on local LLMs (Llama 3.2, Qwen 2.5 via Ollama).
- Implemented a self-repairing coding loop (generate, test, diagnose, fix, commit), Windows .exe packaging, cited web research and PDF question answering with local embeddings.
- Secured it with a Windows AppContainer sandbox, SSRF-safe browser, HMAC-chained audit log and signed phone commands; covered by 365 automated tests.

A.I.R – Artificial Intelligence Rover (Final-Year Project) | *Python, YOLOv8, Ollama, Arduino UNO Q*

- Trained a YOLOv8n object detector on 592 labelled images (mAP@50 0.951, precision 0.914, recall 0.948) and integrated 8 on-device AI models on a Debian-based Arduino UNO Q.
- Wrote the rover's AI "brain": behaviour modes, offline conversation with Qwen 2.5 1.5B (~2.5 s per reply), Moonshine speech-to-text and Piper text-to-speech.
- Built a 2D simulator where the real AI code drives a virtual MCU; fixed an obstacle-avoidance bug, cutting wall collisions to 0 in a 5-minute run.

Multimodal Disaster Response Decision Support | *PyTorch, DistilBERT, ResNet50, PuLP, Streamlit*

- Fine-tuned DistilBERT (15 humanitarian categories) and ResNet50 (damage severity); redesigned the fusion into separate damage and urgency axes after showing naive blending underperformed.
- Allocated rescue teams, ambulances and supplies with fairness-constrained linear programming plus LLM explanations: +24.8% priority-weighted coverage over baseline, under 100 ms per report on CPU.

Ladies Fancy Store – E-commerce Website (live) | *Flask, PostgreSQL, Razorpay, Vercel*

- Built and deployed a production shop for a local business (ladies-fancy-store.vercel.app): catalogue, cart, Razorpay + cash-on-delivery checkout, three delivery modes and an admin panel; auto-deploys from GitHub.

Back Benchers Food Stall – Online Ordering | *Flask, Razorpay, PostgreSQL*

- Online ordering with Razorpay webhook signature verification and server-side pricing; admin board moves orders Confirmed → Preparing → Ready → Completed.

Automation & Vision Tools | *n8n, Airtable, OpenCV, MediaPipe*

- n8n agent that classifies feedback with an LLM, routes it, logs it to Airtable and emails a reply; OpenCV pipeline with adaptive grayscale and histogram analysis; MediaPipe hand-gesture recognizer for 11 robot commands.

INTERNSHIPS

Embedded Systems & IoT Intern – Sreshta Semicon Technologies Pvt. Ltd. **Jun 2026 – Jul 2026**

- Embedded development and IoT architecture project work, connecting hardware sensor data to software.

PCB Design & Electronics Intern – Pantech Solutions **May 2026 – Jun 2026**

EDUCATION

B.Tech, Electronics and Communication Engineering – MVGR College of Engineering (A), Vizianagaram **2023 – 2027**

Intermediate (MPC) – Sri Chaitanya Junior College, Vizianagaram **2021 – 2023**

ACHIEVEMENTS & LEADERSHIP

- Team Lead – ranked 28th of 100+ teams, Autodesk 36-hour hackathon (ANITS, Jan 2026); built the prototype's Fusion 360 CAD.
- SPOC Coordinator – organised an electronics workshop at MVGR's Aadhrita tech fest and mentored 50+ participants.

Languages: English, Telugu, Hindi